

Trainings ▾

General Information

A hydraulic pump can be mounted on the front of the gear housing or rear of the gear housing on the left-bank side of the engine, and on the front of the gear cover on the right-bank side of the engine.

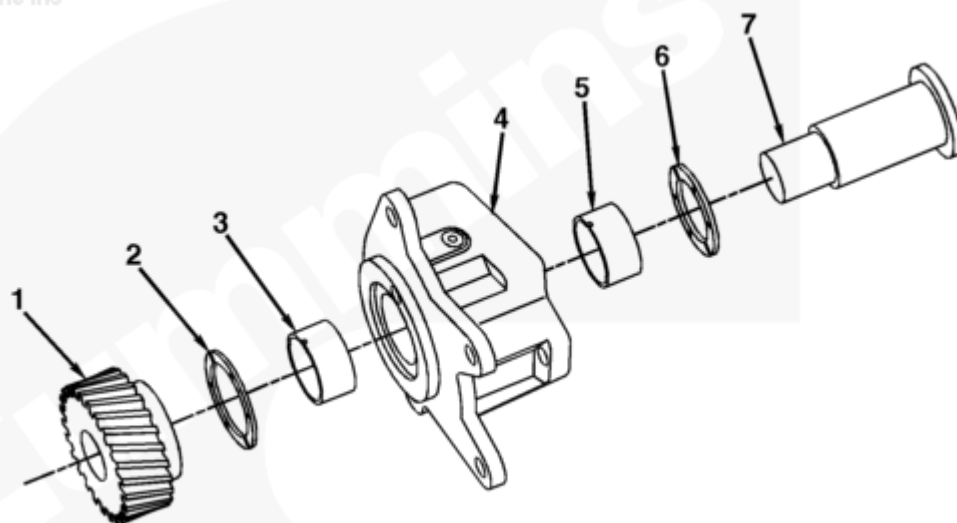
The Society of Automotive Engineering (SAE) A two-bolt-type flange is required to mount a hydraulic pump to the rear of the gear housing on the left-bank side of the engine. The SAE B four-bolt-type flange is required to mount a hydraulic pump to the front gear cover. There is a specific drive required to match the different sizes of spline drives required for optional hydraulic pumps.

Engines that do **not** have a hydraulic pump drive installed **must** have covers installed on the gear cover and housing at the appropriate locations.

A hydraulic pump drive can be installed by removing the appropriate cover and installing the correct drive for the selected mounting location and hydraulic pump.

There are presently three different hydraulic drives available for the three mounting locations. While these drives use different components, their disassembly, inspection, and assembly are the same. The procedures identified will be the same for all three hydraulic drives.

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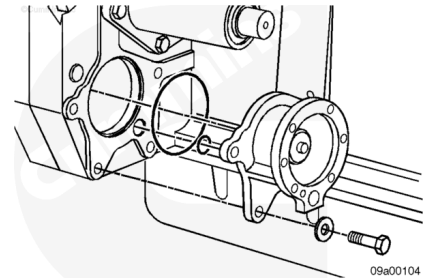
1. Gear, hydraulic pump drive
2. Bearing, thrust
3. Bushing

4. Housing, hydraulic pump drive
5. Bushing
6. Bearing, thrust
7. Driveshaft, hydraulic pump.

Remove

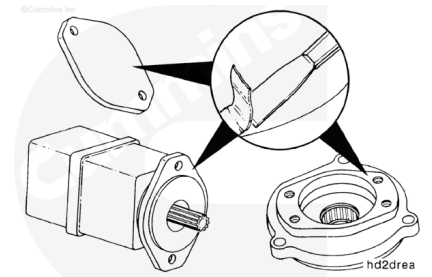
Remove the hydraulic pump.

Remove the capscrews securing the hydraulic pump drive to the gear housing. Remove the drive and discard the o-rings.

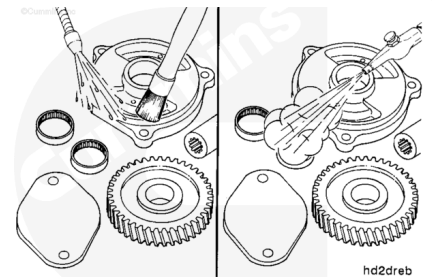


Clean

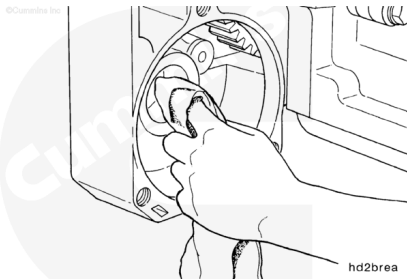
Remove the gasket material from the hydraulic pump adapter and hydraulic pump, if equipped, or cover plate surface.



Clean the hydraulic pump adapter, cover plate, gear, and needle bearings with solvent. Dry with compressed air.



Clean the needle bearing bore in the gear housing with solvent and dry with a clean cloth.



⚠ WARNING ⚠

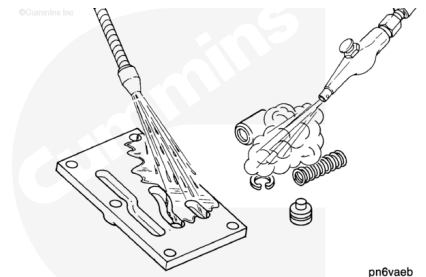
Wear appropriate eye and face protection when using compressed air. Flying debris or dirt can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

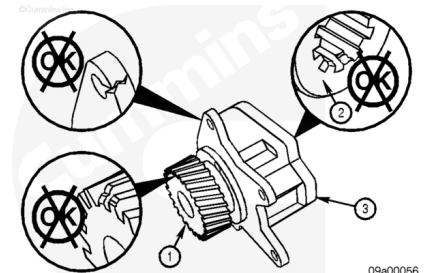
Use solvent to clean the outer housing of the drive.

Dry with compressed air.



Check the following components for damage:

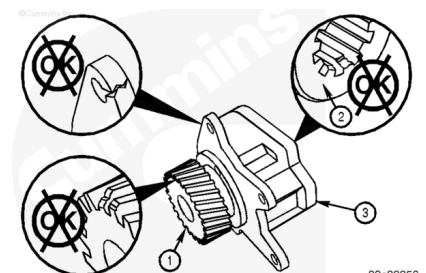
1. Drive Gear
2. Involute Spline (in the end of the shaft)
3. Drive Housing.



Inspect for Reuse

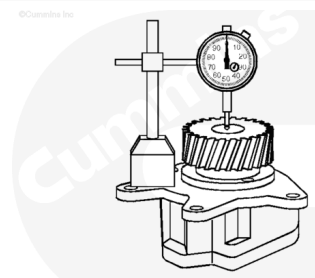
Check the following components for damage:

1. Drive gear
2. Involute spline (in the end of the shaft)
3. Drive housing.



Measure the hydraulic pump drive end clearance.

Hydraulic Pump Drive End Clearance		
mm		in
0.100	MIN	0.004
0.381	MAX	0.015



Use the following procedure if the hydraulic pump drive is out of specification. Refer to Procedure 009-019 in Section 9 (/qs3/pubsys2/xml/en/procedures/57/57-009-019.html) for rebuild.

Install

Install the bore sealing and lubricating oil passage o-rings, capscrews, and hydraulic pump drive onto the gear housing or gear cover.

Torque Value:

Hydraulic Pump Drive

1. 47 n•m [35 ft-lb]

Install the hydraulic pump.

